

DATA ANALYTICS · BUSINESS INTELLIGENCE · DECISION SUPPORT

AIRPORT OPERATIONS DSS

An End-to-End SQL & Power BI Analytics Solution for Airport Operations

MYSQL + POWER BI

01

Project Overview

RAW AIRPORT DATA

Flights · passengers · gates

SQL PROCESSING

ETL · cleaning · relationships

BUSINESS ANALYSIS

KPIs · views · management logic

POWER BI DSS

Interactive decisions

Business Problem & Objectives

How is the airport performing?

Where are operational pressures concentrated?

Which passengers require more assistance?

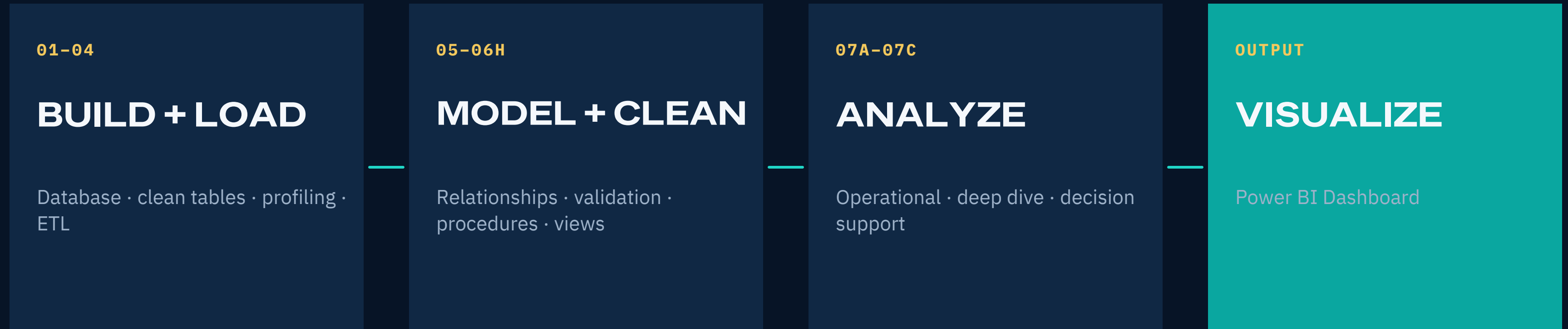
OBJECTIVE / Transform operational data into actionable Decision Support System insights.

Dataset

OCT 2024 — DEC 2024

▪ Flights	▪ Passengers	▪ Baggage	▪ Gate_Events
▪ Security_Screening	▪ Staff_Shifts	▪ Maintenance_Logs	

Data Engineering & SQL Pipeline



Structured, repeatable and traceable from source to decision

Data Cleaning & Validation



Missing values + duplicates



Reconstructed operational timing



Referential integrity + categories



Standardized Boolean + timestamps

Business Analysis Framework

LEVEL 01

Basic Operational Analysis

KPIs · airline volumes · delay performance · passenger workload

LEVEL 02

Operational Deep Dive

Baggage · aircraft utilization · maintenance · security · shifts

LEVEL 03

Decision Support

Passenger impact · benchmarking · risk · assistance · workforce

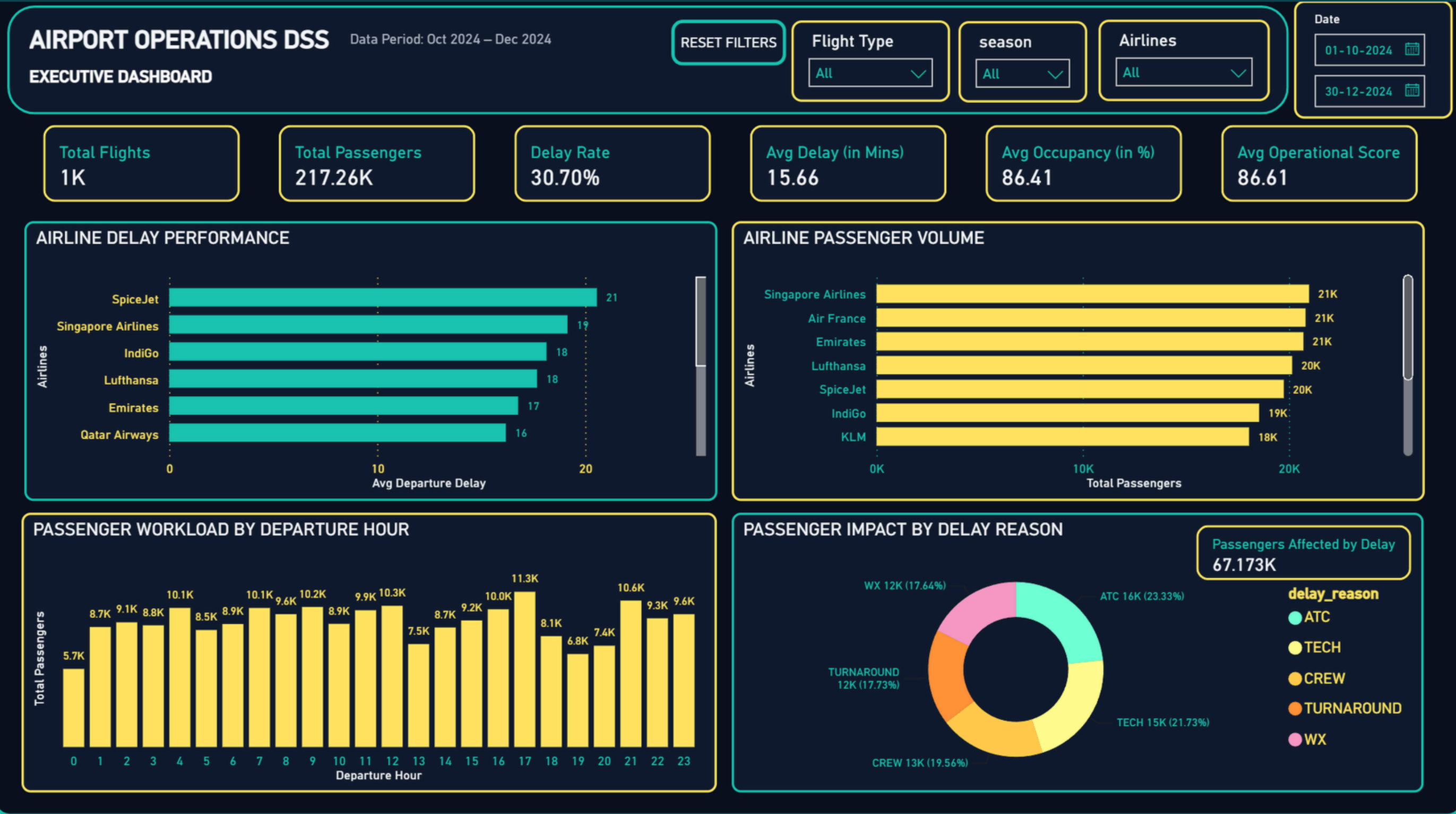
MEASURES → CONTEXT → MANAGEMENT PRIORITIES

Power BI Dashboard Architecture



KPI CARDS · SLICERS · CROSS-FILTERING · PAGE NAVIGATION

Executive Overview



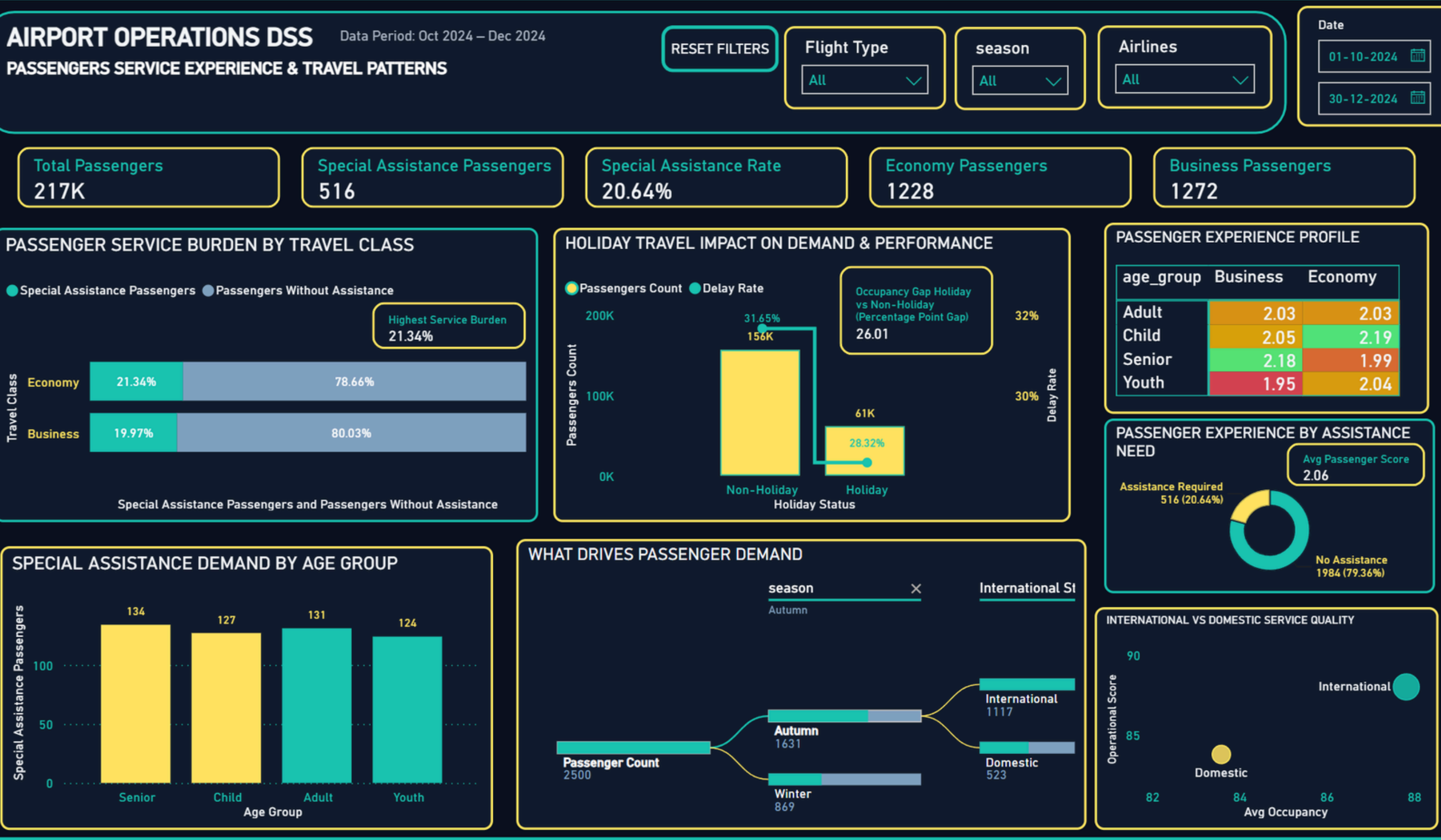
WHAT IS HAPPENING AT THE AIRPORT?

Operational Performance



WHERE DOES MANAGEMENT NEED TO ACT?

Passenger & Service Planning



Key Insights & Business Findings

Delay performance + passenger impact

Identify operational priorities

Airline benchmarking + baggage workload

Support targeted review

Utilization + maintenance severity + inspection risk

Reveal maintenance priorities

Assistance + screening + overtime

Guide service and workforce planning

Business Value / Decision Support

RAW DATA

CLEAN DATA

OPERATIONAL METRICS

BUSINESS ANALYSIS

DECISION SUPPORT

Monitor operational health +
bottlenecks

Compare airlines + understand
passenger impact

Support allocation + targeted
investigation

FROM RAW AIRPORT DATA TO DECISION SUPPORT

Built an end-to-end Airport Operations DSS integrating SQL, ETL, cleaning, analysis and Power BI.

REAL-TIME INTEGRATION • PREDICTIVE DELAYS • DEMAND FORECASTING • PREDICTIVE MAINTENANCE •
AUTOMATED ALERTS • STAFFING OPTIMIZATION